

Teaching Statement

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Teaching Philosophy

My goal as a teacher is to help students move confidently among an economic question, a formal model, and evidence about human behavior. Students should learn not only how to reproduce an argument, but also how to identify its assumptions, understand its intuition, and decide whether it provides a useful account of the problem at hand. This requires rigor without unnecessary opacity. I therefore treat mathematics as a language for expressing economic reasoning rather than as a collection of formulas to be memorized.

Three principles guide my teaching. First, I make the structure of an argument visible. I decompose difficult proofs and derivations into their economic and mathematical steps and show how each step relates to the question being answered. Second, I ask students to reason actively. Before presenting a solution, I invite them to predict an outcome, identify a relevant assumption, or propose the next step of an argument. Third, I connect theoretical benchmarks to data and behavior. An experiment, empirical example, or counterexample can help students understand both the power of a model and the conditions under which its predictions may fail.

These principles reflect the same perspective that motivates my research in behavioral and experimental economics. Formal theory provides disciplined predictions, while observed choices reveal how beliefs, reasoning, and social preferences affect behavior. In the classroom, the gap between the two is not merely a complication. It is an opportunity for students to learn how economists formulate, test, and revise explanations.

Making Formal Reasoning Accessible

I have served as a teaching assistant for undergraduate, master's, doctoral, and DBA courses at Renmin University of China, The Chinese University of Hong Kong, Shenzhen, and the Hong Kong University of Science and Technology. My responsibilities have included independently preparing and leading tutorials, designing tutorial notes and problem sets, developing grading rubrics, grading assignments and examinations, holding office hours, and answering questions through course platforms. These experiences have required me to teach students with widely different levels of mathematical preparation, professional experience, and familiarity with economics.

One particularly formative experience was serving as the teaching assistant for *Advanced Econometrics II* at CUHK Shenzhen while I was a master's student. I led a weekly 90-minute tutorial for doctoral students. Because each part of an advanced econometric argument depends on several earlier results, I prepared a detailed knowledge map for each topic. The maps made explicit how definitions, assumptions, propositions, and examples fit together. During tutorials, I translated compact notation into verbal and economic intuition, worked through representative examples, and then returned to the formal result so that students could see why each assumption was needed.

This approach treats mathematics as a language that students can learn to interpret. Rather than asking them to remember an estimator or theorem as an isolated object, I asked what problem it solves, what information it uses, and what would change if an assumption failed. Examples then allowed students to practice moving in both directions: from formal notation to intuition and from an economic question back to the appropriate formal tool.

I used a related approach when teaching *Microeconomic Analysis* to master's students at HKUST. For each problem, I first identified the economic concept, theorem, or proof technique that the problem was designed to test. I then reconstructed the reasoning before solving the particular exercise. Problems in consumer theory, for example, became opportunities to revisit optimization, comparative statics, and the logic behind the Slutsky decomposition, rather than occasions to manipulate equations mechanically. Students therefore reviewed the course material while learning how to apply it in a new setting.

Several students approached me after tutorials or at the end of the semester to say that the sessions had made difficult material clearer and had contributed substantially to their understanding of the course. Their comments reinforced an important lesson for my teaching: a tutorial should do more than supply answers. It should help students recognize a structure that they can apply independently to the next problem.

Active and Experiential Learning

Even in mathematically demanding courses, I do not want tutorials to be one-way presentations. I regularly pause during a derivation or problem and ask students to identify the relevant result, predict the direction of an effect, or suggest the next step. These questions allow students to uncover gaps in their own reasoning before seeing the completed solution. They also give me immediate information about which parts of an argument require another explanation.

In my future courses, I will extend this practice through structured active-learning activities. In microeconomics and game theory, students can make decisions in short strategic environments before seeing the equilibrium analysis. Comparing their predictions, choices, and stated beliefs makes concepts such as dominance, best responses, Bayesian updating, and equilibrium selection concrete. In econometrics, students can first predict the consequences of a research-design problem and then use simulated or real data to examine those consequences. Short polls, small-group problem solving, and prediction-before-solution activities can make even a large class participatory without sacrificing analytical depth.

My research provides natural classroom environments for this form of learning. A sender–receiver game can introduce information transmission, honesty, trust, and the effects of repeated interaction. A sequential resistance game can help students distinguish a coordination problem from an inference problem. A team-production game can illustrate how effort observability, compensation transparency, and payoff inequality affect incentives. In each case, students can first experience the strategic environment, formulate a theoretical benchmark, and then evaluate whether observed choices support that benchmark.

I would use these examples as entry points rather than as substitutes for the broader literature. My aim is to show students how an economist moves from a substantive question to a model, from a model to testable hypotheses, and from evidence back to an assessment of the explanation. This process connects economic theory, behavioral science, and empirical analysis.

Designing Courses Around Learning

As an instructor, I would begin course design by identifying what students should be able to do at the end of the course. In *Principles of Microeconomics*, for example, students should be able to use incentives, opportunity cost, equilibrium, and welfare analysis to organize a substantive question. In *Intermediate Microeconomics*, they should be able to translate an economic problem into a formal model, derive its implications, and explain the result in economic language. In econometrics, they should be able to move from a research question to an estimand, identify the assumptions required for interpretation, select an appropriate method, conduct relevant statistical inference, and communicate uncertainty.

Lectures, tutorials, assignments, and assessments should all support these objectives. I would combine cumulative problem sets with low-stakes exercises that reveal misunderstandings before a major assessment. Clear grading rubrics would distinguish among correct calculation, valid reasoning, appropriate interpretation, and effective communication. When returning work, I would identify common errors and explain not only why an answer was incorrect, but also which step in the reasoning produced the error.

Project-based learning would allow students to integrate concepts across a course. In an introductory course, students might select a real-world question, identify the incentives and tradeoffs involved, and evaluate a policy proposal using a simple economic framework. In an intermediate course, they could formulate and analyze a model before discussing which predictions are empirically testable. In econometrics, a staged project could require students to define a question, construct or obtain data, pre-specify an analysis, implement it, and communicate the result to both technical and nontechnical audiences.

For quantitative courses, I would integrate mathematical reasoning with data and computation. My teaching experience includes using statistical software demonstrations with DBA students. In future econometrics courses, students could work with real and simulated data to see how an assumption affects an estimate, how sampling variation appears across repeated samples, and how alternative research designs change the interpretation of a result. Computation would not replace theory; it would make the implications of the theory observable.

Inclusive Teaching and Student Support

I have taught students who differ substantially in age, nationality, educational background, mathematical preparation, and professional experience. My classes have included undergraduates encountering econometrics for the first time, master's and doctoral students working through advanced theory, and DBA students applying quantitative methods to research and managerial questions. Except for my teaching at Renmin University, my tutorials have been conducted in English in international learning environments.

These experiences have taught me that students may struggle with the same material for different reasons. One student may understand the economic intuition but find the notation unfamiliar. Another may be comfortable with mathematics but uncertain about interpretation. Others may understand a solution when they see it but not know how to begin independently. I therefore explain important concepts through several mutually reinforcing forms: formal notation, verbal intuition, worked examples, graphical representations, and, where appropriate, computational demonstrations.

Student support also requires attention to the practical constraints that shape students' ability to participate. When supporting a master's-level course at HKUST, I learned that some students were completing internships during the day. I therefore scheduled the tutorial in the evening and recorded each session in full, uploading the recording to the course platform. Students who could not attend could follow the complete discussion, while those who participated could revisit difficult derivations when reviewing the material. This arrangement preserved a common set of learning objectives while providing more than one way for students to engage with the course.

I also respond to students' expressed academic needs. When students ask for more opportunities to practice, I provide additional exercises when feasible or direct them to problems that target the relevant concept. When a question calls for a broader or more advanced treatment, I recommend books and other resources suited to the student's preparation and goals. These requests also provide useful feedback: if several students need the same kind of practice, I can incorporate additional scaffolding into later tutorials or future versions of the course.

More generally, I have posted tutorial notes and recordings on course platforms, provided additional office hours by Zoom when students needed them, and answered questions during demanding examination periods. These practices allow students to revisit difficult material at their own pace and to seek assistance through more than one channel. Responsiveness does not mean lowering expectations or providing a separate standard for each student. It means identifying reasonable barriers to learning and helping students acquire the resources needed to meet shared academic expectations.

As an instructor, I would make this support more systematic. Early low-stakes exercises would help me identify variation in students' preparation. Optional review materials could support students who need additional mathematical or statistical background without slowing the progression of the entire class. Clear rubrics, examples of model reasoning, and discussions of common mistakes would make academic expectations transparent.

I would also vary the form of participation. Full-class discussion can be supplemented by small groups, anonymous polling, brief written responses, and office-hour conversations. Inclusive teaching, in this sense, does not mean reducing intellectual difficulty. It means giving students multiple ways to engage with that difficulty, demonstrate their understanding, and improve after receiving feedback.

Courses, Mentoring, and Curricular Contributions

I am prepared to teach Principles of Microeconomics, Intermediate Microeconomics, Statistics for Economics, Introductory Econometrics, Applied Econometrics, and more advanced courses in econometrics. In my primary fields, I would be enthusiastic about teaching Behavioral Economics, Experimental Economics, Game Theory, Information Economics, Behavioral Game Theory, Political Economy, Economics of Organizations, and Behavioral Public Policy.

I would be especially interested in developing a course in *Behavioral and Experimental Economics*. Students would participate in classroom experiments, derive theoretical benchmarks, analyze behavioral evidence, and design their own experimental projects. The course would emphasize that behavioral economics is not simply a catalogue of departures from standard models. Students would learn to formalize competing explanations and design evidence that can distinguish among them.

A culminating project could ask student teams to transform a substantive question from policy, organizations, health, the environment, or technology into an experimental design. Depending on the level of the course, students could prepare experimental instructions, construct an online prototype, specify treatments and outcome measures, conduct a power analysis, and write a pre-analysis plan. Where actual data collection was not feasible or appropriate, they could analyze simulated or publicly available data constructed around the intended design.

I would also integrate experimental reasoning into econometrics education. Students often learn econometrics primarily through completed observational datasets. A broader approach can teach them to think about how data are generated: how to define an estimand, select outcome measures, design an intervention, anticipate the resulting data structure, determine an appropriate sample size, and conduct a transparent and reproducible analysis. This approach would connect my strengths in econometrics, experimental methods, and formal theory rather than treating them as separate teaching areas.

Beyond the classroom, I would welcome the opportunity to advise undergraduate theses, independent studies, and course-based research projects involving microeconomic theory, game theory, experimental design, behavioral economics, programming, or data analysis. I would help students narrow a broad interest into an answerable question, distinguish a mechanism from a general topic, and develop a feasible research plan. Staged feedback on the question, model or identification strategy, data plan, preliminary analysis, and final presentation would make the research process more manageable and allow students to improve through revision.

My aim is for students to leave my courses with both analytical discipline and intellectual independence. They should be able to use economic models rigorously, evaluate evidence critically, communicate their reasoning clearly, and adapt what they have learned to unfamiliar problems. Whether they continue to academic research or apply economics in other settings, I want them to see theory and evidence not as separate bodies of knowledge, but as complementary tools for understanding economic and strategic behavior.